



TIER 1 — CLOUD COMPUTING

Comprehensive Study Guide, Architectural Comparisons & 25 Solved Practice MCQs

1. What is Cloud Computing? ★★★★★

Definition:

Cloud computing is the on-demand delivery of computing resources such as servers, storage, databases, networking, and software over a network, typically with usage-based pricing.

In simple words:

Instead of buying and maintaining all computer infrastructure yourself, you use computing resources provided by a cloud provider over the Internet.

Traditional IT:

Suppose you want to build a website. You might need:

- Server
- Storage
- Networking
- Database
- Cooling
- Electricity
- Maintenance
- Security

You purchase and maintain everything.

```
graph TD
    A[Your Company] --> B[Own Data Center]
    B --> C[Servers]
```

Cloud Computing:

Instead:

```
graph TD
    A[Your Application] --> B[Internet]
    B --> C[Cloud Provider]
    C --> D[Servers / Storage / Database]
```

You consume the resources you need.

Examples of cloud providers:

- Amazon Web Services (AWS)
- Microsoft Azure
- Google Cloud (GCP)

2. Why Cloud Computing? ★★★★★

Cloud computing provides several important advantages:

1. On-demand resources: You can provision resources when needed.

Need server? → Provision cloud VM → Use it

2. Scalability: You can increase or decrease resources as demand changes.

Example: Normal traffic: **2 servers** | Festival sale: **20 servers**

3. Cost efficiency: Instead of purchasing expensive infrastructure upfront, cloud users can often pay based on consumption/service terms. This can reduce **Capital Expenditure (CapEx)** and shift spending toward **Operating Expenditure (OpEx)**.

4. Global access: Cloud resources can be accessed from different locations over networks.

5. High availability: Cloud architectures can use redundant resources to keep applications available despite failures.

6. Faster deployment: You can provision infrastructure much faster than physically purchasing and installing servers.

3. Cloud Service Models ★★★★★

This is probably the most important cloud MCQ topic. There are three traditional service models:

1. **IaaS** (Infrastructure as a Service)
2. **PaaS** (Platform as a Service)
3. **SaaS** (Software as a Service)

IaaS — Infrastructure as a Service

The provider gives you infrastructure such as virtual machines, storage, networking, and compute resources. You manage much of the software stack yourself.

Example: You rent a virtual machine in the cloud.

Cloud Provider → Physical Hardware → Virtualization → VM
↑
You manage OS + applications

Examples include: AWS EC2, Azure Virtual Machines, Google Compute Engine.

💡 Easy memory: IaaS → Infrastructure

PaaS — Platform as a Service

The cloud provider manages more of the underlying infrastructure and platform. You primarily focus on your application and code.

You → Application / Code → PaaS → Infrastructure managed by provider

Examples: Azure App Service, Google App Engine, AWS Elastic Beanstalk.

💡 Easy memory: PaaS → Platform for developers

SaaS — Software as a Service

A complete software application is provided to the user. You generally use the software rather than manage the underlying infrastructure.

User → Web/App → Complete Software

Examples: Gmail, Microsoft 365, Salesforce.

💡 Easy memory: SaaS → Software ready to use

🔥 IaaS vs PaaS vs SaaS Control Hierarchy

More control
↑
IaaS
PaaS
SaaS
↓
Less management

Responsibility Matrix

Component	IaaS	PaaS	SaaS
Physical hardware	Provider	Provider	Provider
Networking	Mostly provider	Provider	Provider
Virtualization	Provider	Provider	Provider
OS	You	Provider	Provider
Runtime	You	Provider	Provider
Application	You	You	Provider
Data	You	You	Shared / provider-managed

Accenture shortcut:

- VM → **IaaS**
- Deploy code without managing servers → **PaaS**
- Use ready-made application → **SaaS**

4. Cloud Deployment Models ★★★★★

Another very common MCQ area. There are four commonly discussed deployment models:

Public Cloud

Cloud infrastructure is provided for use by multiple customers over a provider's infrastructure.



Examples: AWS, Azure, Google Cloud.

Advantages: Lower upfront infrastructure investment, Highly scalable, Easy provisioning.

Example: A startup hosts its website entirely on AWS.

Private Cloud

Cloud infrastructure is dedicated to a single organization.

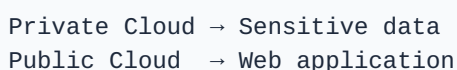


Advantages: Greater control, Custom security/policy requirements, Suitable for regulated environments.

Disadvantage: Can require more infrastructure management and cost.

Hybrid Cloud

Combination of **Private Cloud + Public Cloud**.



A company may keep sensitive workloads in a private environment while using public cloud capacity for scalable applications.

Community Cloud

Infrastructure is shared by organizations with common requirements.

Example: Several organizations in the same regulated industry sharing infrastructure designed around common security/compliance needs (less frequently tested than public/private/hybrid).

Deployment Model Memory:

- **Public** → Provider's shared cloud environment
- **Private** → Dedicated to one organization
- **Hybrid** → Public + Private
- **Community** → Shared by organizations with common requirements

5. Virtualization & 6. Hypervisor ★★★★★

Definition of Virtualization:

Virtualization is the creation of virtual computing resources, such as virtual machines, on physical hardware. One physical server can run multiple virtual machines.

Physical Server → Hypervisor → VM1 | VM2 | VM3

Why virtualization?

- *Without virtualization:* Server 1 → App A, Server 2 → App B, Server 3 → App C (hardware capacity remains unused).
- *With virtualization:* One powerful server → Hypervisor → VM1, VM2, VM3 (resources utilized efficiently).

Hypervisor

A hypervisor is software/firmware that creates and manages virtual machines.

Type 1 — Bare Metal: Runs directly on physical hardware.

Hardware → Hypervisor → VMs

Examples: VMware ESXi, Microsoft Hyper-V, Xen.

Type 2 — Hosted: Runs on top of a host operating system.

Hardware → Host OS → Hypervisor → VM

Examples: Oracle VirtualBox, VMware Workstation.

Type 1 vs Type 2 Comparison

Feature	Type 1 (Bare-Metal)	Type 2 (Hosted)
Location	Directly on hardware	On host OS
Naming	Bare-metal hypervisor	Hosted hypervisor
Environment	Common in enterprise data centers	Common on desktops / labs
Overhead	Generally less overhead	More host-OS overhead

Remember:

- **Type 1** → Hardware first
- **Type 2** → Host OS first

7. Virtual Machine ★★★★★

A Virtual Machine (VM) is a software-defined computer running on a physical host through virtualization. A VM includes Virtual CPU, Virtual RAM, Virtual disk, Virtual network interface, and Guest operating system.

Physical Server → Hypervisor → VM → Linux / Windows → Application

8. Scalability & 9. Elasticity ★★★★★

Scalability: Scalability is the ability of a system to handle increasing workload by adding resources (e.g., 100 users → 10,000 users).

Vertical Scaling (Scale Up): Increase the power of an existing machine (e.g., 2 CPU, 8 GB RAM → 8 CPU, 32 GB RAM). Same server, more resources.

Horizontal Scaling (Scale Out): Add more machines (Server 1, Server 2, Server 3, Server 4) instead of making one server extremely powerful.

Vertical vs Horizontal

Vertical Scaling (Scale Up/Down)	Horizontal Scaling (Scale Out/In)
Increase/decrease single machine resources	Add/remove machines
Limited by physical machine hardware capacity	Can scale across hundreds of machines
Often simpler architecture	Often better suited for distributed systems

🧠 MCQ Shortcut:

- Adding more servers to handle increased traffic → **Horizontal scaling**
- Increasing RAM/CPU of one server → **Vertical scaling**

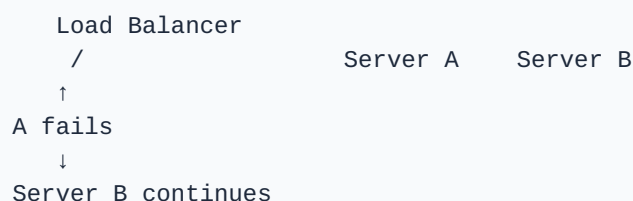
Elasticity: Elasticity is the ability of a cloud system to *dynamically and automatically* add or remove resources in response to changing demand (e.g., 2 servers → traffic increases to 10 servers → traffic drops back to 2 servers).

Scalability vs Elasticity Distinction:

- **Scalability:** Ability to handle growth (capacity planning).
- **Elasticity:** Ability to automatically adjust with dynamic demand spikes.

10. Availability, 11. Reliability & 12. Fault Tolerance ★★★★★

Availability: The degree to which a system remains accessible and operational when needed. Achieved via redundant servers, multiple availability zones, load balancing, failover mechanisms, and health checks.



Reliability: Refers to the ability of a system to consistently perform correctly over time without failure.

- *Available but unreliable:* Website opens, but calculations return incorrect results.
- *Reliable but unavailable:* Works perfectly when running, but frequently offline due to crashes.

Fault Tolerance: The ability of a system to continue operating without interruption despite the failure of one or more components (achieved via active hardware redundancy).

Server A ❌ | Server B ✅ | Server C ✅ → Application continues seamlessly

Fault Tolerance vs High Availability:

- **High Availability:** Minimize downtime.
- **Fault Tolerance:** Zero downtime/interruption during failure.
- *Accenture Rule:* Failure doesn't stop system → **Fault tolerance**

13. CapEx vs OpEx ★★★★★

CapEx (Capital Expenditure): Money spent upfront on purchasing physical assets (servers, storage, networking equipment, physical datacenter building).

OpEx (Operational Expenditure): Ongoing operating costs. Cloud services shift spending toward OpEx based on actual pay-as-you-go consumption.

CapEx (Capital Expenditure)	OpEx (Operational Expenditure)
Upfront capital investment	Ongoing operational spending
Buy physical hardware assets	Consume on-demand cloud services
Physical infrastructure maintenance	Service-based pay-as-you-go model
Higher upfront commitment	Flexible consumption

🧠 **MCQ:** A company stops buying physical servers and instead pays a cloud provider based on consumption
→ **CapEx to OpEx shift**

14–16. Cloud Characteristics & The 5 NIST Essentials ★★★★★

14. Resource Pooling: Cloud providers pool computing resources (CPU, RAM, storage, network) and dynamically allocate them across multi-tenant customers.

15. On-Demand Self-Service: Users provision resources automatically via web consoles or APIs without requiring manual provider intervention.

16. Measured Service: Cloud usage is monitored, measured, and billed transparently based on actual consumption (CPU hours, storage GB, bandwidth).

☁️ The 5 Essential Cloud Characteristics (NIST Model):

1. **On-demand self-service:** Users provision resources themselves.
2. **Broad network access:** Services accessible over standard network mechanisms.
3. **Resource pooling:** Shared multi-tenant resources dynamically managed.
4. **Rapid elasticity:** Provisioned/released automatically based on demand.
5. **Measured service:** Usage monitored and metered for billing.

 Accenture Cloud Computing — Tier 1 MCQs

25 High-Frequency Multiple Choice Questions with Explanations and Shortcuts

Q1. Cloud Computing — Which statement best defines cloud computing?

- A. Buying and maintaining physical servers in your own data center
- B. Delivering computing resources on demand over a network
- C. Using only local computers without Internet connectivity
- D. Storing all data on a personal hard drive

✓ Answer: B. Delivering computing resources on demand over a network

- A  — This is traditional/on-premises infrastructure.
- B  — Cloud computing provides resources such as compute, storage, databases, and networking on demand.
- C  — Cloud services commonly use network connectivity.
- D  — Local storage isn't cloud computing.

 **Remember: Cloud = On-demand resources over a network****Q2. IaaS — A developer rents a virtual machine from a cloud provider and manages the operating system and applications themselves. Which service model is this?**

- A. SaaS
- B. PaaS
- C. IaaS
- D. FaaS

✓ Answer: C. IaaS

- A  SaaS — Complete software is provided to the user.
- B  PaaS — Provider manages the platform/runtime while developer focuses mainly on the application.
- C  IaaS — Provides infrastructure such as VMs, storage, and networking.
- D  FaaS — Function/serverless execution model.

 **VM → IaaS**

Q3. PaaS — A developer wants to deploy application code without managing the underlying operating system, servers, or runtime infrastructure. Which model is most suitable?

- A. IaaS
- B. PaaS
- C. SaaS
- D. Private Cloud

✓ **Answer: B. PaaS**

- A ✗ IaaS — Customer generally manages more of the infrastructure, including the OS.
- B ✓ PaaS — Provider manages underlying platform; developer focuses on application code.
- C ✗ SaaS — User consumes a complete application.
- D ✗ Private Cloud — This is a deployment model, not a service model.

🧠 **Developer + deploy code + less infrastructure management → PaaS**

Q4. SaaS — Which is the best example of SaaS?

- A. Virtual machine
- B. Cloud storage infrastructure
- C. Gmail
- D. Virtual network

✓ **Answer: C. Gmail**

- A ✗ — A VM is typically an IaaS resource.
- B ✗ — Storage services fall under cloud infrastructure offerings.
- C ✓ — Gmail is a complete software application consumed by users.
- D ✗ — Networking infrastructure is not SaaS.

🧠 **Ready-to-use application → SaaS**

Q5. IaaS vs PaaS — Which cloud model generally gives the customer more control over the operating system?

- A. SaaS
- B. PaaS
- C. IaaS
- D. None

✓ **Answer: C. IaaS**

- A ✗ — SaaS hides most infrastructure and platform management.
- B ✗ — PaaS abstracts more of the infrastructure.
- C ✓ — With IaaS, customers generally manage the guest OS.
- D ✗ — IaaS clearly provides this control.

🧠 **More customer control: IaaS → PaaS → SaaS**

Q6. Public Cloud — A startup hosts its application on AWS infrastructure shared among many customers. What type of deployment model is this?

- A. Private Cloud
- B. Public Cloud
- C. Community Cloud
- D. Hybrid Cloud

✓ Answer: B. Public Cloud

- A  — Private cloud is dedicated to one organization.
- B  — Public cloud infrastructure is offered by a cloud provider to multiple customers.
- C  — Community cloud is shared by organizations with common requirements.
- D  — Hybrid combines public and private environments.

 **AWS/Azure/GCP → Usually public cloud**

Q7. Private Cloud — A bank operates a cloud environment dedicated exclusively to its own organization. Which deployment model is this?

- A. Public Cloud
- B. Private Cloud
- C. Hybrid Cloud
- D. Community Cloud

✓ Answer: B. Private Cloud

- A  — Public cloud serves multiple customers.
- B  — Infrastructure is dedicated to one organization.
- C  — No public-cloud component is mentioned.
- D  — Community cloud involves multiple organizations with shared requirements.

Q8. Hybrid Cloud — A company keeps sensitive workloads in its private cloud but uses AWS for its scalable web application. What model is this?

- A. Public Cloud
- B. Private Cloud
- C. Hybrid Cloud
- D. Community Cloud

✓ Answer: C. Hybrid Cloud

- A  — Not entirely public.
- B  — Not entirely private.
- C  — Combines private and public cloud environments.
- D  — No shared community arrangement is described.

 **Private + Public = Hybrid**

Q9. Virtualization — What is the primary purpose of virtualization?

- A. Convert domain names to IP addresses
- B. Create virtual computing resources on physical hardware
- C. Encrypt every network packet
- D. Replace the Internet

✓ Answer: B. Create virtual computing resources on physical hardware

- A  — DNS performs name resolution.
- B  — Virtualization allows physical hardware to host virtual machines/resources.
- C  — Encryption is a security function.
- D  — Virtualization doesn't replace the Internet.

Q10. Hypervisor — Which software manages virtual machines and provides them access to physical hardware resources?

- A. Compiler
- B. Hypervisor
- C. Firewall
- D. DNS server

✓ Answer: B. Hypervisor

- A  Compiler — Converts source code into machine code.
- B  Hypervisor — Creates/manages VMs and allocates hardware resources.
- C  Firewall — Controls network traffic.
- D  DNS server — Resolves domain names.

 **VM manager → Hypervisor**

Q11. Type 1 Hypervisor — Which statement describes a Type 1 hypervisor?

- A. It runs directly on physical hardware
- B. It always runs inside a browser
- C. It requires Windows as the host OS
- D. It is an application running on top of a host OS

✓ Answer: A. It runs directly on physical hardware

- A  — Type 1 is a bare-metal hypervisor.
- B  — Not browser-based.
- C  — Not Windows-specific.
- D  — That's the Type 2 model.

 **Type 1 → Hardware → Hypervisor → VM**

Q12. Type 2 Hypervisor — Which architecture represents a Type 2 hypervisor?

- A. Hardware → Hypervisor → VM
- B. Hardware → Host OS → Hypervisor → VM
- C. Hardware → VM → Hypervisor
- D. VM → Hardware → Hypervisor

✓ **Answer: B. Hardware → Host OS → Hypervisor → VM**

- A ✗ — Type 1.
- B ✓ — Type 2 runs on a host operating system.
- C ✗ — Incorrect architecture.
- D ✗ — Incorrect order.

🧠 **Type 2 → Host OS first**

Q13. Vertical Scaling — A company increases a server from 4 CPU + 8 GB RAM to 16 CPU + 32 GB RAM. What type of scaling is this?

- A. Horizontal scaling
- B. Vertical scaling
- C. Geographic scaling
- D. Multicast scaling

✓ **Answer: B. Vertical scaling**

- A ✗ — Horizontal scaling means adding more machines.
- B ✓ — Increasing resources of the same machine.
- C ✗ — Not a standard scaling classification here.
- D ✗ — Multicast is a networking communication method.

🧠 **Bigger machine → Vertical / Scale Up**

Q14. Horizontal Scaling — An e-commerce company increases its application servers from 3 to 10 during a sale. What type of scaling is this?

- A. Vertical scaling
- B. Horizontal scaling
- C. Static scaling
- D. Manual routing

✓ **Answer: B. Horizontal scaling**

- A ✗ — Would increase CPU/RAM of existing servers.
- B ✓ — Adds more servers.
- C ✗ — Not the relevant concept.
- D ✗ — Routing has nothing to do with scaling here.

🧠 **More machines → Horizontal / Scale Out**

Q15. Elasticity — Traffic suddenly increases from 1,000 to 100,000 users. The cloud platform automatically adds servers and removes them when traffic falls. This is an example of:

- A. Virtualization only
- B. Elasticity
- C. Static routing
- D. Data encryption

✓ **Answer: B. Elasticity**

- A ✗ — Virtualization may be involved but doesn't describe automatic resource adjustment.
- B ✓ — Resources dynamically increase/decrease with demand.
- C ✗ — Routing is unrelated.
- D ✗ — No encryption is being described.

🧠 **Demand changes → Resources automatically adjust → Elasticity**

Q16. Scalability vs Elasticity — Which statement is correct?

- A. Scalability means the ability to handle increased workload; elasticity means dynamically adjusting resources with demand
- B. Scalability and elasticity are exactly the same concept
- C. Elasticity refers only to data encryption
- D. Scalability applies only to physical servers

✓ **Answer: A**

- A ✓ Correct.
- B ✗ — Closely related but not identical.
- C ✗ — Elasticity concerns resource adjustment.
- D ✗ — Scalability applies to cloud and distributed systems too.

🧠 **Remember: Scalability = Can handle growth | Elasticity = Dynamically adjusts**

Q17. Availability — A cloud application continues to serve users even when one server fails because another server takes over. Which concept is primarily demonstrated?

- A. High availability
- B. Data compression
- C. DNS resolution
- D. Virtualization only

✓ **Answer: A. High availability**

- A ✓ — Redundancy reduces service downtime.
- B ✗ — Compression reduces data size.
- C ✗ — DNS resolves names.
- D ✗ — Virtualization may be part of the architecture but isn't the main concept.

🧠 **Keep service running → High Availability**

Q18. Fault Tolerance — A system continues operating even after one of its hardware components fails. Which concept best describes this?

- A. Fault tolerance
- B. SaaS
- C. DNS
- D. Horizontal routing

✓ **Answer: A. Fault tolerance**

- A ✓ — Designed to continue despite component failures.
- B ✗ — SaaS is a service model.
- C ✗ — DNS is name resolution.
- D ✗ — Not a standard concept.

🧠 **Failure occurs but system continues → Fault tolerance**

Q19. Reliability — Which term refers primarily to a system's ability to perform correctly and consistently over time?

- A. Reliability
- B. Latency
- C. Scalability
- D. Virtualization

✓ **Answer: A. Reliability**

- A ✓ — Consistent correct operation.
- B ✗ — Latency is delay.
- C ✗ — Scalability concerns handling increased workload.
- D ✗ — Virtualization creates virtual resources.

Q20. CapEx — A company purchases physical servers, storage systems, and networking equipment for its own data center. This is primarily:

- A. OpEx
- B. CapEx
- C. SaaS
- D. Elasticity

✓ **Answer: B. CapEx**

- A ✗ OpEx — Ongoing operational spending.
- B ✓ CapEx — Capital investment in physical assets.
- C ✗ SaaS — Software service model.
- D ✗ Elasticity — Dynamic resource adjustment.

🧠 **Buy physical infrastructure → CapEx**

Q21. OpEx — A company stops purchasing physical servers and instead pays a cloud provider for resources according to its service/usage model. This generally shifts spending toward:

- A. CapEx
- B. OpEx
- C. Hardware depreciation only
- D. Physical networking

✓ **Answer: B. OpEx**

- A ✗ — CapEx is upfront capital investment.
- B ✓ — Cloud consumption commonly shifts spending toward operational expenditure.
- C ✗ — Hardware depreciation applies to CapEx.
- D ✗ — Physical networking is avoided in standard cloud hosting.

🧠 **Consume cloud services → generally OpEx**

Q22. Cloud Characteristics — Which of the following is NOT one of the commonly recognized five essential characteristics of cloud computing?

- A. On-demand self-service
- B. Resource pooling
- C. Rapid elasticity
- D. Manual hardware installation for every user

✓ Answer: D. Manual hardware installation for every user

The five commonly taught characteristics are:

1. On-demand self-service
2. Broad network access
3. Resource pooling
4. Rapid elasticity
5. Measured service

- A ✓ — Cloud characteristic.
- B ✓ — Cloud characteristic.
- C ✓ — Cloud characteristic.
- D ✗ — Cloud aims to eliminate manual physical provisioning for users.

Q23. Resource Pooling — A cloud provider dynamically allocates computing resources from a shared pool to different customers. Which cloud characteristic is this?

- A. Resource pooling
- B. Static routing
- C. Encryption
- D. Fault injection

✓ Answer: A. Resource pooling

- A ✓ — Provider pools resources and allocates them dynamically.
- B ✗ — Routing concept.
- C ✗ — Security mechanism.
- D ✗ — Testing/failure-injection technique.

Q24. Measured Service — A cloud provider tracks how much storage, compute, or network traffic a customer consumes for monitoring and billing. Which cloud characteristic does this represent?

- A. Measured service
- B. Private cloud
- C. Vertical scaling
- D. Hypervisor

✓ **Answer: A. Measured service**

- A ✓ — Cloud usage is monitored/measured.
- B ✗ — Deployment model.
- C ✗ — Scaling technique.
- D ✗ — VM management software.

Q25. 🔥 Mixed Scenario — A company wants to deploy an application quickly. Its requirements are:

- It doesn't want to manage physical servers.
- It doesn't want to manage the operating system.
- Developers want to focus primarily on application code.
- The application should automatically scale with demand.

Which combination is most appropriate?

- A. IaaS + manual scaling
- B. PaaS + elasticity
- C. Private cloud + physical servers
- D. On-premises hardware + manual scaling

✓ **Answer: B. PaaS + elasticity**

- A ✗ — IaaS provides more infrastructure responsibility and manual scaling doesn't satisfy the requirement as well.
- B ✓ — PaaS reduces infrastructure/OS management, while elasticity allows resources to dynamically adjust.
- C ✗ — Requires more infrastructure management.
- D ✗ — Contradicts the goal of avoiding physical-server management.